

# Kirill Mazur

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## Education

### Imperial College London

Ph.D. student in Robotic Vision

2021 - present

- Dyson Robotics Lab; Supervisor: [Andrew J. Davison](#)

### Skolkovo Institute of Science and Technology, GPA: 3.83 out of 4.00

2018 - 2020

M.Sc. in Mathematics and Computer Science (with honors)

- Thesis: “Cloud Transformers”. Supervisors: [Victor Lempitsky](#) and [Gonzalo Ferrer](#)
- First year project: “Learnable point cloud descriptors for depth-only odometry and SLAM”

### Yandex School of Data Analysis (YSDA), GPA: 4.84 out of 5.00

2016 - 2018

Computer Science (Masters-level degree school organized by Yandex)

### Higher School of Economics (HSE), GPA: 9.22 out of 10.0

2014 - 2018

B.Sc. in Mathematics

- Thesis: “On the Families of Quartics and Rational Curves on the Quotient of the Quartic by the Involution”

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## Publications

### 4D Primitive-Mâché: Glueing Primitives for Persistent 4D Scene Reconstruction

[Kirill Mazur](#), Marwan Taher, Andrew J. Davison

[[project page](#), [video](#)]

Primitive-based 4D replayable reconstructions from casual videos.

(CVPR) *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026

### KV-Tracker: Real-Time Pose Tracking with Transformers

Marwan Taher, Ignacio Alzugaray, [Kirill Mazur](#), Xin Kong, Andrew J. Davison

[[project page](#), [video](#)]

Real-time pose tracking from feed-forward reconstruction models.

(CVPR) *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026

### SuperPrimitive: Scene Reconstruction at a Primitive Level

[Kirill Mazur](#), Gwangbin Bae, Andrew J. Davison

[[project page](#), [video](#), [code](#)]

Introduces a new representation for monocular scene reconstruction based on image segments.

(CVPR) *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024

**Honorable Mention** at CVPR Demo Track

### Feature-Realistic Neural Fusion for Real-Time, Open Set Scene Understanding

[Kirill Mazur](#), Edgar Sucar, Andrew J. Davison

[[project page](#), [video](#)]

Presents a new real-time high-dimensional feature fusion technique for open set scene understanding.

(ICRA) *IEEE International Conference on Robotics and Automation* 2023

### Point-Based Clothing Modeling

Ilya Zakharkin\*, [Kirill Mazur](#)\*, Artur Grigorev, Victor Lempitsky

[[project page](#), [code](#)]

Presents a new technique for a visual try-on and clothes re-targeting for complex garments.

(ICCV) *IEEE/CVF International Conference on Computer Vision* 2021

### Cloud Transformers: A Universal Approach To Point Cloud Processing Tasks

[Kirill Mazur](#), Victor Lempitsky

[[project page](#), [code](#)]

Presents a new layer for 3D point clouds processing that achieved SoTA results on four various tasks.

(ICCV) *IEEE/CVF International Conference on Computer Vision* 2021

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## Experience

### Meta

Aug. 2024 - Feb. 2025

### AI Research Scientist Intern at [Project Aria](#)

- Developed a model for dense metric geometry estimation for Aria device
- Large-scale training of dense point-map estimators, including data preparation and model iteration

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| <b>Samsung AI Center</b><br><b>Research Engineer at Virtual Human Telepresence Lab</b> <ul style="list-style-type: none"> <li>Published 2 first (co-)author papers at a top computer vision venue (ICCV).</li> <li>“Cloud Transformers”: Research on neural point cloud processing.</li> <li>“Point-Based Clothing Modeling”: Research on neural 3D clothes modeling and its visual try-on.</li> </ul> | Jun. 2019 - Jul. 2021    |
| <b>Yandex</b><br><b>Intern ML Engineer at Computer Vision Lab</b> <ul style="list-style-type: none"> <li>Developed a model for Object Localization which reduced the response time by 75% and doubled a target product metric</li> <li>Adapted Tensorflow’s object detection project for Yandex Infrastructure</li> </ul>                                                                              | Jun. 2018 - Sep. 2018    |
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| <b>Invited Talks</b>                                                                                                                                                                                                                                                                                                                                                                                   |                          |
| <b>Technical University of Munich (TUM)</b><br>Week-long research visit to <b>Daniel Cremers’</b> lab                                                                                                                                                                                                                                                                                                  | July. 2024               |
| <b>Massachusetts Institute of Technology (MIT)</b><br>“Always-On Single-Image Priors for Spatial AI”                                                                                                                                                                                                                                                                                                   | Jun. 2024                |
| <b>Rank Symposium: Neural Rendering in Computer Vision</b><br>“Feature-Realistic SLAM”                                                                                                                                                                                                                                                                                                                 | Aug. 2022                |
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| <b>Academic Service</b>                                                                                                                                                                                                                                                                                                                                                                                |                          |
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| <b>Reviewer</b><br>WACV 2025, ICCV 2025, CVPR 2024-2025, ICRA 2023-2024                                                                                                                                                                                                                                                                                                                                |                          |
| <b>Imperial College London</b><br>TA at Robotics course                                                                                                                                                                                                                                                                                                                                                | Spring 2023, Spring 2024 |
| <b>Skolkovo Institute of Science and Technology</b><br>TA at Deep Learning course                                                                                                                                                                                                                                                                                                                      | Spring 2020              |
| <b>Higher School of Economics / Yandex School of Data Analysis</b><br>Research seminar curator                                                                                                                                                                                                                                                                                                         | Sep. 2018 - Dec. 2019    |
| <b>Yandex School of Data Analysis (YSDA)</b><br>TA at Algorithms and Data Structures course                                                                                                                                                                                                                                                                                                            | Sep. 2018 - Jan. 2019    |